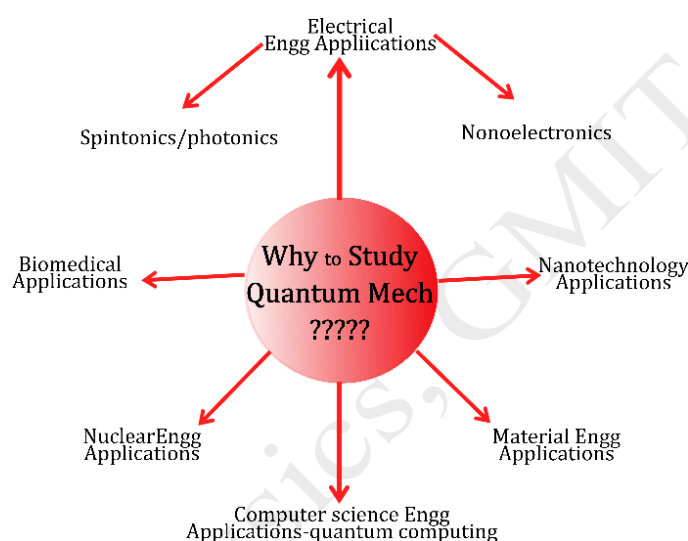


Quantum Mechanics

Syllabus: de Broglie Hypothesis and Matter Waves, de Broglie wavelength and derivation of expression by analogy, Phase Velocity and Group Velocity, Heisenberg's Uncertainty Principle and its application (Non-existence of electron inside the nucleus - Non Relativistic), Principle of Complementarity, Wave Function, Time independent Schrödinger wave equation (Derivation), Physical Significance of a wave function and Born Interpretation, Expectation value, Eigen functions and Eigen Values, Particle inside one dimensional infinite potential well, Quantization of Energy States, Waveforms and Probabilities.

Numerical problems- on de Broglie Hypothesis, Heisenberg's Uncertainty Principle, Energy Eigen Values for a particle in 1D infinite potential well



History: At the end of 19th century, scientists felt that all the laws of physics (which were known at that period of time) are enough to explain all the events occur in nature. It was believed that there are only two kinds in nature. First the bodies which are made up of particles and second radiation. All particles obey Newton's laws of motion and radiation obey Maxwell's equations of electromagnetism. These laws are now a days known as **classical physics**.

Fortunately, at the same time some experiments were performed and the experimental results were not in comparison with so called classical theory. Some of them are **Black body radiation**, Photo electric effect and Compton Effect etc. In order to overcome the drawbacks of Classical Physics, a new types of laws of physics have been introduced which are now known as **Quantum Physics**. Black body radiation is one of the great effects which has led to the concept of Modern Physics/Quantum Physics.

Need of Quantum Mechanics: One of the interesting areas of research in Physics during latter part of 19th century was about emission of radiation from **black body** (It is an object, which absorbs all radiations incident on it and emit those radiations on heating). Theoretical investigations were individually carried out by **Wien** (in German) and **Rayleigh & Jeans** (in England) to account for the energy emitted by black body and partially succeeded. While, the equation given by **Wien** was fitted well with experimental curve at shorter wavelengths but failed at longer side. Conversely, **Rayleigh-Jeans** formula was fitted well with experimental curve at longer wavelengths and failed at shorter side. Finally, **Max Planck** worked out an empirical equation that fitted the entire curve completely.

Following the success, **Planck** analyzed how his equation could be interpreted and realized that energy emitted by black body was quantized. Earlier, it was thought as continuous as predicted by traditional mechanics (i.e., classical mechanics), $\therefore$ failed to explain experimental observations. This quantization signified the need for new mechanics (i.e., Quantum Mechanics). Thus, it was with **Planck's Eqn.**, for energy density of the black body radiation, quantum mechanics was originated.

de Broglie Hypothesis:

The origin of quantization of energy lies in the dual nature behavior observed in particle and wave aspects. For example, Compton observed that X-rays are scattered like particle by an electron and Bragg observed that X-rays get diffracted by a crystal like a wave. By observing such a dual nature of light, **Louis de Broglie** in 1924 put forward a bold hypothesis that:

Statement: “Just like radiation (light) which behaves as a wave and as a particle at different circumstances, similarly matter entities (say electrons) which ordinarily behaves as particle can exhibit wave properties under appropriate circumstances”.

Matter (de Broglie) waves: are the waves associated with material particles in motion.

When a particle has a momentum p and its motion is associated with a wavelength λ , called de Broglie wavelength, then

$$\lambda = \frac{h}{p} \quad \left| \text{where, } h \text{ is Planck's constant} \right.$$

Max Karl Planck (1858-1947)

- German Physicist
- **Known for:** Planck constant, Planck postulate, Planck's law of black body radiation, Fokker–Planck equation, Nernst–Planck equation, Third law of thermodynamics, etc.,
- **Awards:** Nobel prize in 1918
- **Doctoral advisors:** G Kirchhoff, H von Helmholtz, Alexander von Brill
- **His notable students:** G L Hertz, Max von Laue, Walter Schottky, Walther Meissner, etc.,

Experimental proof-Wave nature of electron:- Davisson & Germer of USA (in 1927) accidentally conducted an experiment to show diffraction of accelerated electrons from nickel crystal just like the wave. This diffraction of electrons confirms their wave nature.

Derivation of expression for de Broglie wavelength (λ) by analogy:

The concept of matter waves is well understood by combining Planck's quantum theory and Einstein's theory. Consider a photon of energy ' E ', frequency ' ν ' moving with velocity ' c ' & wavelength ' λ '

By Einstein's mass-energy relation, $E = mc^2$ (1)

By Planck's theory of radiation, $E = h\nu$ (2)

Equating eqn. (1) and (2), we have, $mc^2 = h\nu \Rightarrow mc^2 = h \frac{c}{\lambda} \quad \left| \because \nu = \frac{c}{\lambda} \right.$

Or, $\lambda = \frac{h}{mc}$

But, $p = mc$, $\therefore \lambda = \frac{h}{p}$

This is the expression for de Broglie wavelength.

If a particle of mass ' m ' moving with velocity ' v ' and momentum ' P ', then according to de Broglie hypothesis the wavelength is given by $\lambda = \frac{h}{mv} = \frac{h}{P}$

Different forms of de Broglie Wavelength (λ):

(i) de Broglie wavelength of an electron accelerated under Potential (V):

Proof: We know that, the energy of an electron accelerated under a potential difference V is given by ' eV '. If ' m ' is the mass and ' v ' is the velocity of electron then the energy equation for non-relativistic case is given by,

$$eV = \frac{1}{2}mv^2 \quad (1)$$

If P is the momentum of the electron, then, $P = mv$

$$\text{Squaring on both sides, } P^2 = m^2v^2 \Rightarrow mv^2 = \frac{P^2}{m} \quad (2)$$

$$\text{Put eqn. (2) in (1), we have, } eV = \frac{P^2}{2m} \Rightarrow P = \sqrt{2meV} \quad (3)$$

From de Broglie hypothesis, we know that, $\lambda = \frac{h}{p}$

$$\therefore \lambda = \frac{h}{\sqrt{2meV}} \quad |\because \text{ using eqn. (3)}|$$

This is the equation for wavelength accelerated electron under the potential V .

Since ' h ', ' m ' & ' e ' are constants, i.e., $h = 6.626 \times 10^{-34}$ JS, $m = 9.1 \times 10^{-31}$ kg & $e = 1.6 \times 10^{-19}$ C, substituting these values in above eqn. we get,

$$\lambda = \frac{1.226}{\sqrt{V}} \quad nm$$

Note: Instead of electron, if a particle of charge ' q ' is accelerated under potential V , then $\lambda = \frac{h}{\sqrt{2mqV}}$

(ii) de Broglie wavelength (λ) in terms of kinetic energy (E):

Since, wkt: $eV = \frac{1}{2}mv^2 \Rightarrow eV = E$

$$\text{Substituting in, } \lambda = \frac{h}{\sqrt{2meV}} \Rightarrow \lambda = \frac{h}{\sqrt{2mE}}$$

Wave Packet:

Defn: A **wave packet** refers to the case where two (or more) waves exist simultaneously.

A wave packet is often referred to as a wave group. This situation is permitted by the principle of superposition.

In physics, a wave packet is a short 'envelop' of localized waves that travel as a group. A wave packet can be analyzed into, or can be synthesized from group of sinusoidal waves of different wave numbers, phase and amplitudes such that they interfere constructively or destructively in the space.

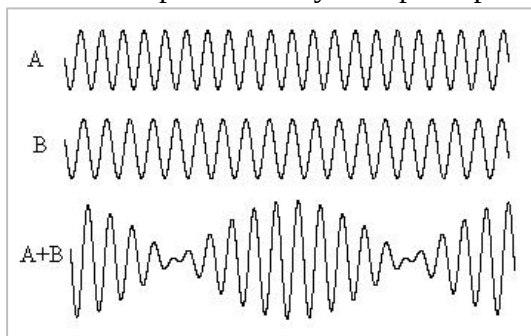
Wave (Phase) Velocity (V_p): The velocity with which individual wave travels in group of waves (Wave packet) is called **phase velocity**.

The **phase velocity** is given by ; $V_p = v\lambda = \frac{\omega}{k}$

Where, ω is the angular frequency, $k = \frac{2\pi}{\lambda}$ is the propagation wave vector (wave number).

Louis de Broglie (1892-1987)

- French Physicist
- **Known for:** Wave nature of electrons, de Broglie-Bohm theory, de Broglie wavelength etc.
- **Awards:** Nobel Prize in Physics (1929)
- **Doctoral advisors:** Paul Langevin



Group Velocity (V_g): The velocity with which a wave packet (group of waves) travels is called **group velocity**. It is usually different from individual phase velocity.

The **group velocity** is given by ; $V_g = \frac{\Delta\omega}{\Delta k} = \frac{d\omega}{dk}$

- The relation between **group velocity** and **phase velocity** is proportionate.

Relation between group velocity and phase velocity

Group velocity is given by $V_g = \frac{d\omega}{dk}$ -----(1)

Phase velocity is given by $V_p = \frac{\omega}{k}$ -----(2)

From equation 2 , $\omega = k V_p$ ----- (3)

Substituting equation 3 in 1, we get

$$V_g = \frac{d\omega}{dk} = \frac{d(V_p k)}{dk} = V_p + k \frac{dV_p}{dk} \text{ -----(4)}$$

But $k = \frac{2\pi}{\lambda}$ -----(5)

By differentiating with respect to λ

$$dk = -\frac{2\pi}{\lambda^2} d\lambda \text{ ----- (6)}$$

Substituting equation 5 and 6 in 4

$$V_g = V_p - \lambda \frac{dV_p}{d\lambda}$$

This gives relation between V_g and V_p

- Schrodinger postulated that a moving particle (e-, etc) is equivalent to a wave packet rather than a single wave. Hence, group velocity = particle velocity.

Principle of Complementarity: In 1928, Niels Bohr told this principle of complementarity.

Principle: According to this, all the objects in atomic dimension requires description of both wave and particle properties for complete knowledge about it, however particle and wave behavior are mutually exclusive. This dualism in quantum system is principle of complementarity.

The complementarity principle holds that objects have certain pairs of complementary properties which cannot all be observed or measured simultaneously.

It Correlates de Broglie hypothesis, Heisenberg's Uncertainty principle and wave packet.

Heisenberg's Uncertainty Principle:

State:-"The position and momentum of a particle cannot be determined accurately and simultaneously. The product of uncertainty in the measurement of position and momentum is always greater than or equal to $\frac{\hbar}{2}$ (where, $\hbar = \frac{h}{2\pi}$)".

$$\text{i.e.,} \quad \Delta x \cdot \Delta p \geq \frac{\hbar}{4\pi}$$

where, Δp and Δx are the uncertainties in the measured values of the momentum and position of the particle, respectively.

Similar to **Position - Momentum** uncertainty, there are other two uncertainty relations also present. They are: **Angular momentum - Angular displacement** uncertainty

$$\text{i.e.,} \quad \Delta L \cdot \Delta \theta \geq \frac{\hbar}{4\pi}$$

and **Energy - Time** uncertainty

$$\text{i.e.,} \quad \Delta E \cdot \Delta t \geq \frac{\hbar}{4\pi}$$

Werner Heisenberg (1901-1976)

- German Physicist
- Known for:** Uncertainty principle, Heisenberg ferromagnet, Heisenberg's microscope, Kramers-Heisenberg formula, etc.
- Awards:** Nobel Prize in Physics (1932)
- Doctoral advisors:** Arnold Sommerfeld, Niels Bohr, Max Born

Application of Uncertainty Principle:

Q: State and explain Heisenberg's Uncertainty principle. Show that the electron emitted during β -decay does not pre-exist inside the nucleus using uncertainty principle.

Non-existence of electron in the nucleus (Non-relativistic case): This is related to the β -ray emission in nuclear physics. We know that the β -rays are actually streams of negatively charged particles, having kinetic energies in the range **3 to 4 MeV**.

“Is it possible that electrons exist inside the nucleus with certain energy.....!????”.

It can be answered by applying Uncertainty Principle as follows:

Proof: The energy of a body of mass m , moving with momentum P can be expressed as,

$$E = \frac{p^2}{2m} \quad (1)$$

and by Uncertainty Principle, $\Delta p_x \cdot \Delta x \geq \frac{h}{4\pi}$

$$\Rightarrow \Delta p_x \geq \frac{h}{4\pi \Delta x} \quad (2)$$

We know that the diameter of nucleus is of the order of 10^{-14} m . Thus, if an electron is to exist inside the nucleus, then the uncertainty in its position Δx must not exceed 10^{-14} m .

$$\text{i.e.,} \quad \Delta x \leq 10^{-14} \text{ m}$$

Using this in **eqn. (2)**, we have,

$$\Delta p_x \geq \frac{6.626 \times 10^{-34}}{4 \times 3.142 \times 10^{-14}}$$

$$\text{or,} \quad \Delta p_x \geq 0.5 \times 10^{-20} \text{ N-s} \quad (3)$$

This is the uncertainty in the momentum of the electron. Now, by making use of this inequality (i.e., **eqn. 3**) in **(1)**, we can say that, in order that electron may exist inside the nucleus, its energy E must be such that,

$$E \geq \frac{(0.5 \times 10^{-20})^2}{2(9.1 \times 10^{-31})} \quad \text{[using eqn. (1) and } m = 9.1 \times 10^{-31} \text{ kg}]$$

$$\Rightarrow E \geq 1.372 \times 10^{-11} \text{ J}$$

$$\text{or,} \quad E \geq 1.372 \times 6.24 \times 10^{18} \text{ eV} \quad \because 1 \text{ J} = \frac{1}{1.6 \times 10^{-19}} \text{ eV}$$

$$\therefore E \geq 85 \text{ MeV}$$

It indicates that if electron exists inside the nucleus, then its energy must be greater than **85 MeV**. But, the experimental investigations on β -decay reveal that, the kinetic energy of β -particle is of the order of **3 to 4 MeV**.

This clearly indicates that, the emitted electrons cannot be the ones, which existed inside the nucleus.

Wave Function:

In quantum mechanics, it is postulated that every particle is associated with a function determined by the physical situation. This function is called '**wave function**'. It contains all possible information about the system. Hence, wave function is also called as **state function**.

The wave function may vary with respect to both **space (x)** and **time (t)**. It is then called total wave function and it is represented by capital Greek letter Ψ (psi). If the wave function varies only with respect to position then it is represented by lowercase Greek letter ' ψ '.

The total wave function can be represented as: $\Psi = Ae^{i(kx-\omega t)}$

where ' A ' is constant (i.e., amplitude), ' ω ' is the angular frequency, ' k ' is propagation wave vector.

Separating the space and time dependent parts: $\Psi = Ae^{ikx} e^{-i\omega t}$

$\Rightarrow \Psi = \psi e^{-i\omega t}$ $|\psi = Ae^{ikx}$ Space dependent wave function

In quantum mechanics, the **wave function** actually accounts for the wave like properties of a particle. It is obtained by solving fundamental eqn. called **Schrodinger Equation**.

Schrodinger Equation:

It can be setup in two different contexts. One, which is general and takes care of both **space** and the **time** variations of the **wave function**, is called **time dependent Schrodinger's equation**. The other one is applicable to steady state conditions in which **wave function** is only the function of **space** not the time, is called **time independent Schrodinger's equation**.

Time independent Schrodinger equation:

Q: Setup 1-dimensional time independent Schrodinger wave equation. (6M)

According to de Broglie theory, for a particle of mass ' m ' moving with a momentum ' P ', associated with a wave of wavelength,

$$\lambda = \frac{h}{p} \Rightarrow P = \frac{h}{\lambda} \quad (1)$$

The wave function describing the de Broglie wave travelling in positive ' x ' direction (i.e., 1-D) can be written using complex notation as,

$$\Psi = A e^{i(kx-\omega t)} \quad (2)$$

where, Ψ is the total wave function, A is a constant and ω is the angular frequency of the wave. Separating the space dependent part in **eqn. (2)**, we can write,

$\Rightarrow \Psi = \psi e^{-i\omega t}$ $(3) \because \psi = Ae^{ikx}$ space-dependent wave function

On differentiating **eqn. (3)** twice wrt ' x ', we get,

$$\frac{d^2\Psi}{dx^2} = e^{-i\omega t} \frac{d^2\psi}{dx^2} \quad (4)$$

Now again differentiating **eqn. (3)** twice wrt ' t ', we get,

$$\frac{d^2\Psi}{dt^2} = -\omega^2 e^{-i\omega t} \psi \quad (5)$$

We have equation for a travelling wave moving with velocity 'c' and displacement 'y' as,

$$\frac{d^2 y}{dx^2} = \frac{1}{c^2} \frac{d^2 y}{dt^2}$$

Similarly, we can write the wave eqn. for de Broglie wave moving with velocity 'c' and displacement Ψ , as

$$\frac{d^2 \Psi}{dx^2} = \frac{1}{c^2} \frac{d^2 \Psi}{dt^2} \quad (6)$$

Substituting eqns. (4) & (5) in (6), we get,

$$e^{-i\omega t} \frac{d^2 \psi}{dx^2} = \frac{1}{c^2} (-\omega^2 e^{-i\omega t}) \psi \quad \text{or,} \quad \frac{d^2 \psi}{dx^2} = \frac{-\omega^2}{c^2} \psi$$

If λ and ν are the wavelength and frequencies of the wave then $\omega = 2\pi\nu$ and $c = \nu\lambda$. Substituting for ' ω ' and ' c ' then above eqn. becomes,

$$\frac{d^2 \psi}{dx^2} = \frac{-4\pi^2 \psi}{\lambda^2} \Rightarrow \frac{1}{\lambda^2} = -\frac{1}{4\pi^2 \psi} \frac{d^2 \psi}{dx^2} \quad (7)$$

For a particle of mass 'm' moving with velocity 'v', the kinetic energy (KE) is;

$$KE = \frac{1}{2} m v^2 = \frac{p^2}{2m}$$

But, from eqn. (1), we know that, $P = \frac{h}{\lambda}$ then above eqn. becomes,

$$KE = \frac{h^2}{2m} \frac{1}{\lambda^2} \quad (8)$$

Substitute the value of $\frac{1}{\lambda^2}$ from eqn. (7) in (8), we get,

$$KE = \frac{-h^2}{8\pi^2 m} \frac{1}{\psi} \frac{d^2 \psi}{dx^2} \quad (9)$$

If the particle's potential energy is 'V', then the total energy (E) becomes,

$$E = KE + V$$

$$\therefore E = \frac{-h^2}{8\pi^2 m} \frac{1}{\psi} \frac{d^2 \psi}{dx^2} + V \quad | \because \text{Using eqn. (9)}$$

$$\text{or,} \quad \frac{-h^2}{8\pi^2 m} \frac{1}{\psi} \frac{d^2 \psi}{dx^2} = E - V$$

$$\text{or,} \quad \frac{d^2 \psi}{dx^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

This is the **time-independent Schrödinger equation** in 1-Dimension.

**Erwin
Schrödinger
(1887-1961)**

- Austrian Physicist
- **Known for:** Schrodinger equation, Schrodinger's cat, Schrodinger's picture, Entropy, quantum biology, Quantum superposition, etc.,
- **Awards:** Nobel Prize in Physics (1933)
- **Doctoral advisors:** Friedrich Hasenöhl

Note: **Time-independent Schrödinger equation** in 3-Dimension:

$$\left(\frac{d^2 \psi}{dx^2} + \frac{d^2 \psi}{dy^2} + \frac{d^2 \psi}{dz^2} \right) + \frac{8\pi^2 m}{h^2} (E - V) \psi(x, y, z) = 0$$

Physical significance of Wave function & its Properties:

Q: What are the properties of wave function? Give the qualitative explanation of Max Born's interpretation. (6M)

Probability density: If Ψ is the wave function associated with the particle, then $|\Psi|^2$ is the probability per unit volume or probability density that a particle will be found at the given region.

Max Born interpretation: If τ be volume of certain region, inside which a particle is known to be present, but where exactly is not known (see **Fig.**). Then the probability of finding the particle in certain element of volume $d\tau$ of τ is always equal to " $|\Psi|^2 d\tau$ ". For this reason $|\Psi|^2$ is called as **probability density** or **probability function**.

Since, probability of occurrence of any event is a real and positive quantity. However, Ψ is complex & has no considerable physical significance. Hence, in order to get real and positive value of $|\Psi|^2$, the wave function is multiplied with its complex conjugate. The product $\Psi \Psi^*$ is always a positive real quantity.

$$\therefore |\Psi|^2 = \Psi \Psi^*$$

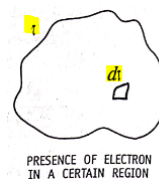
Normalization / Expectation Value: If we are certain that, the particle is definitely present inside the volume τ then, as per the statistical rule,

$$\int_0^\tau |\Psi|^2 d\tau = 1$$

Here, the value 1 for probability means, it is clear that particle is certainly present in the volume τ . If we are not yet all certain about presence of particle in a finite volume then, the 'particle must exist somewhere in the space'.

$$\therefore \int_{-\infty}^{\infty} |\Psi|^2 d\tau = 1$$

This interpretation was first given by **Max Born** in 1926.



Properties of Wave function:

- Ψ is **single valued** everywhere.
- Ψ is **finite** everywhere.
- Ψ and its **first derivative** with respect to its variable are **continuous**.
- For bound states, Ψ must **vanish at infinity**, then $\Psi \Psi^*$ must also vanish at infinity.

Eigen Functions & Eigen values:

Defⁿ: "The wave functions that possess all the 4 properties of a wave function are named in quantum mechanics as **Eigen functions**".

Operator: It is a mathematical entity, which converts one function/state to other function/state.

The information of a wave function Ψ can be extracted by performing certain mathematical operations on Ψ .

Eigen values: When the operators are applied on Eigen function and produce Eigen function back along with a constant multiplied it then such constant values obtained for a physical observable are called as **Eigen values**.

i.e., if an operator $\hat{A}$ acts on eigen function Ψ and gives back Ψ along with Eigen value ' λ ', then

$$\hat{A}\Psi = \lambda \Psi$$

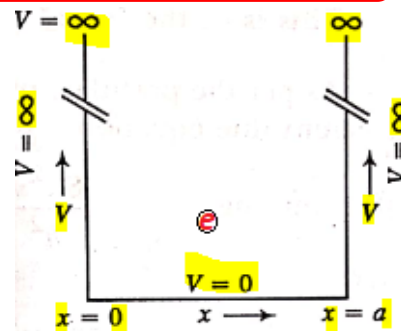
$$\text{Eg:- (i) } \frac{d}{dx} (e^{ax}) = a e^{ax},$$

$$(ii) \frac{d^2}{dx^2} (\sin 2x) = -4 (\sin 2x), \text{ etc}$$

Application of Schrödinger wave equation: Particle (electron) in 1-D potential well of infinite height:-

Q: Starting from Schrodinger's time independent wave equation, derive the expression for energy Eigen value and Eigen function for an electron present in 1-D potential well of infinite height. (08M)

Consider a particle of mass ' m ' is free to move in the X-direction only from region $x = 0$ to $x = a$ as shown in **Fig.** In this region, its potential is $V = 0$. Outside this region, the potential is $V = \infty$. Such configuration of potential is called as **infinite potential well**.



Outside the well, the Schrödinger equation is given by,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}(E - \infty)\psi = 0 \quad (1) \quad | \text{ Since, } V = \infty,$$

The above **eqn.** holds good only if $\psi = 0$.

$\therefore$ the only possible solution is $\psi = 0$ at the walls and also everywhere outside the well. (**Condⁿ-I**)

$\Rightarrow |\psi|^2 = 0$ It indicates that particle cannot be found at all outside the well.

Inside the well, the Schrödinger equation is given by,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}E\psi = 0 \quad (2) \quad | \text{ Since, } V = 0$$

In **eqn. (2)**, putting $\frac{8\pi^2m}{h^2}E = K^2$ (3)

We have, $\frac{d^2\psi}{dx^2} + K^2\psi = 0$

The solution of the above equation is given by,

$$\psi = C \cos Kx + D \sin Kx \quad (4)$$

But, $\psi = 0$ at $x = 0$ | As per **Condⁿ-I**

$\therefore$ **eqn. (4)** becomes, $0 = C \cos K \cdot 0 + D \sin K \cdot 0$

$\Rightarrow C = 0$ (5)

Again, $\psi = 0$ at $x = a$ | As per **Condⁿ-I**

$\therefore$ **eqn. (4)** becomes, $0 = C \cos Ka + D \sin Ka$

$\Rightarrow D \sin Ka = 0$ | $\because C = 0$, using **eqn. (5)**

Here, $D \neq 0$ $\therefore \sin Ka = 0$

$\Rightarrow Ka = n\pi$

$\Rightarrow K = \frac{n\pi}{a}$ (6)

where, $n = 0, 1, 2, 3, 4, \dots$ is known as **quantum number**.

Substituting the values of **eqn. (5)** and **(6)** in **eqn. (4)**, we get,

$$\psi = D \sin \frac{n\pi}{a} x \quad (7)$$

And, substituting the value of **eqn. (6)** in **eqn. (3)**, we get,

$$\frac{8\pi^2 m}{h^2} E = \left(\frac{n\pi}{a} \right)^2$$

$$\Rightarrow E = \frac{n^2 h^2}{8ma^2} \quad (8)$$

This is the required **Energy Eigen value equation** for the electron in 1-D potential well of infinite height. Since, $n = 0, 1, 2, 3, 4, \dots$ then corresponding values that E takes are the quantized energy Eigen values $E_1, E_2, E_3, E_4, \dots$

In **eqn. (7)**, the value $n = 0$ is not acceptable as $\psi = 0$ at $n = 0$ by eqn. (7). Hence the lowest value for n is 1. Consequently, the lowest allowed energy corresponding to $n = 1$ is called as '**zero point energy**' or **ground state energy**. The energy states corresponding to $n > 1$ are known as **excited states energies**.

To find D in eqn. (7): Consider the normalized wave function for the particle from $x = 0$ to a , as,

$$\int_0^a |\Psi_n|^2 d\tau = 1$$

Substituting Ψ from **eqn. (7)**, we have,

$$\int_0^a D^2 \sin^2 \left(\frac{n\pi}{a} \right) x dx = 1$$

$$\Rightarrow D^2 \left[\int_0^a \frac{1}{2} [1 - \cos 2 \left(\frac{n\pi}{a} \right) x] dx \right] = 1 \quad | \because \sin^2 \theta = \frac{1}{2} [1 - \cos 2\theta]$$

$$\text{i.e.,} \quad \frac{D^2}{2} \left[\int_0^a dx - \int_0^a \cos \left(\frac{2n\pi}{a} \right) x dx \right] = 1$$

$$\text{or,} \quad \frac{D^2}{2} \left[[x]_0^a - \frac{a}{2n\pi} \sin \frac{2n\pi x}{a} \Big|_0^a \right] = 1$$

$$\Rightarrow \frac{D^2}{2} \left[a - \frac{a}{2n\pi} \sin 2n\pi \right] = 1$$

$$\text{or,} \quad \frac{D^2}{2} [a] = 1 \quad | \because \sin 2n\pi = 0$$

$$\therefore D = \sqrt{\frac{2}{a}}$$

Putting the value of ' D ' in **eqn. (7)**, we get,

$$\psi_n = \sqrt{\frac{2}{a}} \sin \left(\frac{n\pi}{a} \right) x$$

This is the **Eigen function** for a particle in an infinite potential well.

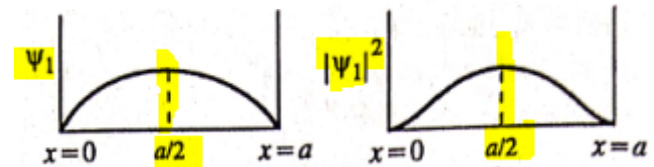
Eigen Functions, Probability densities and Energy Eigen values for a particle in Potential well:

Case (i): $n = 1$:- This is the **ground state** and the particle is normally found in this state.

Eigen function becomes; $\psi_1 = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi}{a}\right)x$

$$\psi_1 = 0 \text{ at } x = 0 \text{ and } x = a$$

But, $\psi_1 = \text{max}$ at $x = \frac{a}{2}$



$|\psi_1|^2$ plot indicates that probability of finding the particle is maximum at the center i.e., $x = \frac{a}{2}$.

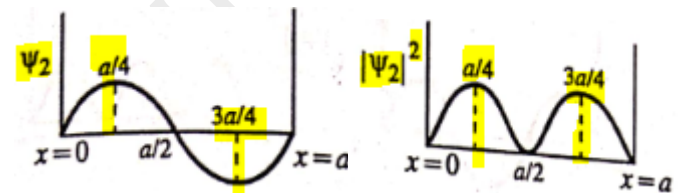
The ground state energy for $n=1$ will be; $E_1 = \frac{h^2}{8ma^2} = E_0$ [for, n^{th} state, $E_n = n^2 E_0$]

Case (ii): $n = 2$:- This is the **first excited state**

Eigen function becomes: $\psi_2 = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi}{a}\right)x$

$$\psi_2 = 0 \text{ at } x = 0, \frac{a}{2}, \text{ and } a$$

But, $\psi_2 = \text{max}$ at $x = \frac{a}{4}$ and $\frac{3a}{4}$



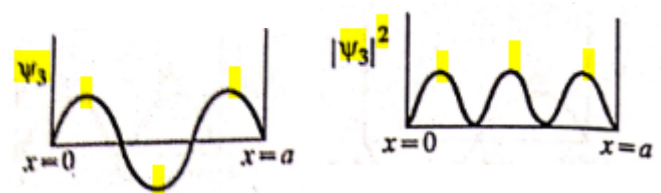
The first excited state energy for $n=2$ will be; $E_2 = \frac{4h^2}{8ma^2}$ or $E_2 = 4E_0$

Case (iii): $n = 3$:- This is the **second excited state**

Eigen function becomes: $\psi_3 = \sqrt{\frac{2}{a}} \sin\left(\frac{3\pi}{a}\right)x$

$$\psi_3 = 0 \text{ at } x = 0, \frac{a}{3}, \frac{2a}{3} \text{ and } a$$

But, $\psi_3 = \text{max}$ at $x = \frac{a}{6}, \frac{a}{2}$ and $\frac{5a}{6}$



The second excited state energy for $n=3$ will be; $E_3 = \frac{9h^2}{8ma^2}$ or $E_3 = 9E_0$



Solved Examples

Example 1. Calculate the de Broglie wavelength associated with an electron having a kinetic energy of 100 eV.

Data: Kinetic energy of the electron, $E=100$ eV

To find : de Broglie wavelength of electron, $\lambda = ?$

Solution: We have the equation,

$$\lambda = \frac{h}{\sqrt{2mE}}$$

From data, $E= 100$ eV= $100 \times 1.602 \times 10^{-19}$ J

Substituting the values for various parameters,

$$\lambda = \frac{h}{\sqrt{2 \times 9.12 \times 10^{-31} \times 100 \times 1.602 \times 10^{-19}}}$$

$$\lambda = 1.227 \times 10^{-10} \text{ m}$$

Example 2. Calculate the momentum of an electron and the de Broglie wavelength associated with it if its KE is 1.5 keV.

Data: Kinetic energy of electron, $E = 1.5$ keV

To find : Momentum of the electron $p = ?$

De Broglie wavelength of electron, $\lambda = ?$

Solution: We have the de Broglie relation,

$$\lambda = \frac{h}{p} \quad (1)$$

We also have the relation between the kinetic energy and de Broglie wavelength,

$$\lambda = \frac{h}{\sqrt{2mE}} \quad (2)$$

Comparing the equations (1) and (2), we have,

$$\frac{h}{p} = \frac{h}{\sqrt{2mE}} \quad \text{or,} \quad p = \sqrt{2mE} \quad (3)$$

From data, $E= 1.5 \times 10^3$ eV = $1.5 \times 10^3 (1.6 \times 10^{-19})$ J

$$\text{or,} \quad E = 2.4 \times 10^{-16} \text{ J}$$

Substituting this value in eqn. (3), we have

$$p = \sqrt{2 \times 9.11 \times 10^{-31} \times 2.4 \times 10^{-16}}$$

$$p = 2.09 \times 10^{-23} \text{ kgms}^{-1}$$

Further, substituting the above in eqn. (1),

$$\lambda = \frac{6.63 \times 10^{-34}}{2.09 \times 10^{-23}}$$

$$\lambda = 3.17 \times 10^{-11} \text{ m}$$

Example 3. The velocity of an electron of a hydrogen atom in the ground state is 2.19×10^6 m/sec. Calculate the wavelength of the de Broglie waves associated with its motion.

Data: Velocity of electron, $v=2.19 \times 10^6$ m/s

To find : de Broglie wavelength, $\lambda = ?$

Solution: de Broglie wavelength of electron is;

$$\lambda = \frac{h}{p} \quad \text{or} \quad \lambda = \frac{h}{mv}$$

$$\lambda = \frac{6.62 \times 10^{-34}}{(9.11 \times 10^{-31})(2.19 \times 10^6)}$$

$$\lambda = 3.31 \times 10^{-10} \text{ m} \quad \text{or,} \quad \lambda = 3.17 \text{ \AA}$$

Example 4. Compute the de Broglie wavelength for a neutron moving with one tenth part of the velocity of light, given, mass of neutron = 1.674×10^{-27} kg.

Data: Mass of neutron $m = 1.674 \times 10^{-27}$ kg

Velocity of neutron; $v = \frac{1}{10} c = 3 \times 10^7$ m/sec

To find : de Broglie wavelength; $\lambda = ?$

Solution: de Broglie wavelength for neutron is;

$$\lambda = \frac{h}{p} \quad \text{or} \quad \lambda = \frac{h}{mv}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{(1.074 \times 10^{-27}) \times (3 \times 10^7)}$$

$$\lambda = 1.314 \times 10^{-14} \text{ m}$$

Example 5. Estimate the potential difference through which a proton is needed to be accelerated so that its de Broglie wavelength becomes equal to $1 \text{ \AA}$. Given that its mass is 1.673×10^{-27} kg.

Data: de Broglie wavelength attained, $\lambda = 10^{-10}$ m

Mass of the proton, $m = 1.673 \times 10^{-27}$ kg

To find : Potential difference needed, $V = ?$

Solution: We have the equation,

$$\lambda = \frac{h}{\sqrt{2meV}} \quad \text{or,} \quad V = \frac{h^2}{2me\lambda^2}$$

$$V = \frac{(6.63 \times 10^{-34})^2}{2 \times 1.673 \times 10^{-27} \times 1.6 \times 10^{-19} \times (10^{-10})^2}$$

$$V = 0.082 \text{ V}$$

Example 6. Evaluate de Broglie wavelength of He nucleus that is accelerated through 500 V. Given mass of proton=mass of neutron= 1.67×10^{-27} kg (May 2022).

Data: Mass of proton=Mass of neutron= 1.67×10^{-27} kg

Potential difference $V = 500$ V

To find : de Broglie wavelength; $\lambda = ?$

Solution: 'He' nucleus has 2 proton & 2 neutrons.

$\therefore$ mass of nucleus, $m = 4m' = 4 \times 1.67 \times 10^{-27}$ kg

Since there are 2 protons, the charge on nucleus,

$$e = 2 \times q = 2 \times (1.6 \times 10^{-19})$$

We have the equation for de Broglie wavelength,

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times (4 \times 1.67 \times 10^{-27}) (2 \times 1.6 \times 10^{-19}) \times 500}}$$

$$\lambda = 4.53 \times 10^{-13} \text{ m}$$

Example 7. A particle having mass of $0.5 \text{ MeV}/c^2$ has a kinetic energy of 100 eV. Calculate the de-broglie wavelength, where c is velocity of light. (Model QP1)

Data: Mass of the particle,

$$m = 0.5 \text{ MeV}/c^2 = \frac{0.5 \times 10^6 \times 1.6 \times 10^{-19}}{(3 \times 10^8)^2} = 8.9 \times 10^{-31} \text{ kg}$$

Kinetic Energy, $E = 100 \text{ eV} = 100 \times 1.6 \times 10^{-19} \text{ J}$

To find : De Broglie wavelength of electron, $\lambda = ?$

Solution: From de- Broglie wavelength equation,

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 8.9 \times 10^{-31} \times 100 \times 1.6 \times 10^{-19}}}$$

$$\lambda = 1.24 \times 10^{-10} \text{ m}$$

Example 8. The KE of an electron is equal to the energy of a photon with a wavelength of 560 nm. Calculate the de-broglie wavelength of the electron. (Model QP2)

Data: Energy of photon is given by,

$$E = \frac{hc}{\lambda} = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{560 \times 10^{-9}} = 0.0354 \times 10^{-17} \text{ J}$$

To find: de-broglie wavelength of electron, $\lambda = ?$

Solution: The de Broglie wavelength of an electron is;

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 0.0354 \times 10^{-17}}}$$

$$\lambda = 8.26 \times 10^{-10} \text{ m}$$

Heisenberg's Uncertainty Principle

Example 9. Infer the Heisenberg's Uncertainty Principle for quantum and classical mechanical measurements.

Solution: Let us consider the following example to infer this (i.e., uncertainties for microscopic and macroscopic systems).

Let us estimate the uncertainty in the position of;

(a) A neutron moving at velocity 5×10^6 m/s (Quantum mechanical measurement).

(b) A 50 kg person moving at velocity 2 m/s (Classical mechanical measurement).

Given: Mass of neutron is 1.67×10^{-27} kg, $h = 6.623 \times 10^{-34}$ J s.

The Heisenberg's Uncertainty in position and momentum is;

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \geq \frac{h}{4\pi\Delta p} \quad \text{or} \quad \Delta x \geq \frac{h}{4\pi mv}$$

(a) For quantum mechanical measurement;

$$m = 1.67 \times 10^{-27} \text{ kg}, \quad v = 5 \times 10^6 \text{ m/s}$$

$$\therefore \Delta x \geq \frac{6.623 \times 10^{-34}}{4 \times 3.142 \times 1.67 \times 10^{-27} \times 5 \times 10^6}$$

$$\Delta x \geq 6.4 \times 10^{-15} \text{ m}$$

This distance is comparable to the size of the nucleus; hence, it can be accepted.

(b) For classical mechanical measurement;

$$m = 50 \text{ kg}, \quad v = 2 \text{ m/s}$$

$$\Delta x \geq \frac{6.623 \times 10^{-34}}{4 \times 3.142 \times 50 \times 2}$$

$$\Delta x \geq 5.26 \times 10^{-37} \text{ m}$$

An uncertainty of this magnitude is beyond human detection; therefore, it can be neglected.

Conclusion: From above inference we conclude that, uncertainties are important for microscopic system, but negligible for macroscopic system.

Example 10. The position and momentum of an electron with energy 0.5 keV are determined. What is the minimum percentage of uncertainty in its momentum if the uncertainty in measurement of its position is 0.5Å? (May 2022)

Data: Energy of electron,

$$E = 0.5 \times 10^3 \text{ eV} = 0.5 \times 10^3 \times 1.6 \times 10^{-19} \text{ J}$$

Maximum uncertainty in position,

$$\Delta x = 0.5\text{\AA} = 0.5 \times 10^{-10} \text{ m}$$

To find: Minimum percentage of uncertainty in momentum = ?

Solution: We have the uncertainty principle as,

$$\Delta x \Delta p \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta p \geq \frac{h}{4\pi\Delta x}$$

$$\rightarrow \Delta p \geq \frac{6.623 \times 10^{-34}}{4 \times 3.142 \times 0.5 \times 10^{-10}}$$

$$\Delta p \geq 1.06 \times 10^{-24}$$

$\therefore$ Uncertainty in momentum is $1.06 \times 10^{-24} \text{ m/s}$

We have the equation for momentum,

$$p = \sqrt{2mE}$$

$$p = \sqrt{2 \times 9.1 \times 10^{-31} \times 0.5 \times 10^3 \times 1.6 \times 10^{-19}}$$

$$\therefore p = 1.207 \times 10^{-23} \text{ Ns}$$

Now,

$$\% \text{ of uncertainty in momentum} = \frac{\Delta p}{p} \times 100$$

$$= \frac{1.06 \times 10^{-24}}{1.207 \times 10^{-23}} \times 100$$

$$= 8.8\%$$

Example 11. In a measurement of position and momentum that involved an uncertainty of 0.003%, the speed of an electron was found to be 800 m/s. Calculate the corresponding uncertainty that arises in determining its position.

Data: Speed of electron, $v = 800 \text{ m/s}$

Uncertainty in measurement of $v = 0.003\%$

To find: Uncertainty in the measurement of position of electron, $\Delta x = ?$

Solution: Uncertainty in speed,

$$\Delta v = 800 \times \frac{0.003}{100} = 0.024 \text{ m/s}$$

& uncertainty in momentum,

$$\Delta p = m \times \Delta v = 9.1 \times 10^{-31} \times 0.024$$

$$= 2.184 \times 10^{-32} \text{ kg m/s}$$

As per Heisenberg's uncertainty principle,

$$\Delta x \geq \frac{h}{4\pi\Delta p}$$

$$\Delta x \geq \frac{6.623 \times 10^{-34}}{4 \times 3.142 \times 2.184 \times 10^{-32}}$$

$$\Delta x \geq 2.4 \times 10^{-3} \text{ m}$$

Example 12. In a measurement of position and velocity of an electron moving with a speed of $6 \times 10^5 \text{ m/s}$, calculate the highest accuracy with which its position could be

determined if the inherent error in the measurement of its velocity is 0.01% for the speed stated.

Data: Speed of electron, $v = 6 \times 10^5$ m/s

Uncertainty involved in measurement of $v = 0.01\%$

To find: Best accuracy with which the position could be measured, $\Delta x = ?$

Solution: Uncertainty in speed,

$$\Delta v = 6 \times 10^5 \times \frac{0.01}{100} = 60 \text{ m/s}$$

From Heisenberg's Uncertainty principle,

$$\Delta x \geq \frac{h}{4\pi m \Delta v}$$

$$\Delta x \geq \frac{6.623 \times 10^{-34}}{4 \times 3.142 \times 9.1 \times 10^{-31} \times 60}$$

$$\Delta x \geq 9.65 \times 10^{-7} \text{ m}$$

Example 13. The inherent uncertainty in the measurement of time spent by Iridium-191 nuclei in the excited state is found to be 1.4×10^{-10} s. Estimate the uncertainty that results in its energy a in the excited state.

Data: Uncertainty in life time in the excited state, $\Delta t = 1.4 \times 10^{-10}$ s

To find: Uncertainty in the energy of the nuclei in the excited state, $\Delta E = ?$

Solution: We have the equation for uncertainty in energy and time as,

$$\Delta E \Delta t \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta E \geq \frac{h}{4\pi \Delta t}$$

$$\Delta E \geq \frac{6.623 \times 10^{-34}}{4 \times 3.142 \times 1.4 \times 10^{-10}}$$

$$\Delta E \geq 3.77 \times 10^{-25} \text{ J} \geq \frac{3.77 \times 10^{-25}}{1.6 \times 10^{-19}}$$

$$\Delta E \geq 2.35 \times 10^6 \text{ eV} \geq 2.35 \text{ MeV}$$

Example 14. A spectral line of wavelength $5461 \text{ \AA}$ has a width of 10^{-4} . Evaluate the minimum time spent by an electron in the upper energy state.

Data: Wavelength, $\lambda = 5461 \times 10^{-10} \text{ m}$

Width of the spectral line (also means uncertainty in the wavelength value),

$$\Delta \lambda = 5461 \times 10^{-10} \times 10^{-4} \approx 10^{-14}$$

To find: The minimum time spent by the electrons in the upper state, $= ?$

Solution: We have equation, $E = h\nu = \frac{hc}{\lambda}$

$$\Delta E = hc \Delta \left(\frac{1}{\lambda} \right) = hc \left(\frac{\Delta \lambda}{\lambda^2} \right) \quad (1)$$

(Ignore negative sign)

As per uncertainty principle,

$$\Delta E \Delta t \geq \frac{h}{4\pi} \rightarrow \Delta t \geq \frac{h}{4\pi \Delta E} \quad (2)$$

Using eqⁿ (1), the eqⁿ (2) becomes,

$$\Delta t \geq \frac{h\lambda^2}{4\pi hc \Delta \lambda} = \frac{\lambda^2}{4\pi c \Delta \lambda}$$

$$\Delta t \geq \frac{(5461 \times 10^{-10})^2}{4 \times 3.142 \times 3 \times 10^8 \times 10^{-14}}$$

$$\Delta t \geq 0.8 \times 10^{-8} \text{ sec}$$

Example 15. The speed of electron is measured to within an uncertainty of $2 \times 10^4 \text{ ms}^{-1}$ in one dimension. What is the minimum width required by the electron to be confined in an atom? (Model QP2)

Solution: From Heisenberg's uncertainty principle,

$$\Delta x \Delta P_x \geq \frac{h}{4\pi}$$

$$\text{But, } P = mv \rightarrow \Delta P_x = m \Delta v$$

$$\therefore \Delta P_x \geq \frac{h}{4\pi m \Delta v}$$

$$\geq \frac{6.625 \times 10^{-34}}{4\pi \times 9.1 \times 10^{-31} \times 2 \times 10^4}$$

$$\Delta x = 0.028 \times 10^{-7} \text{ m}$$

$$\Delta x = 2.8 \times 10^{-9} \text{ m} = 2.8 \text{ nm}$$

Energy Eigen value of Particle in 1-D**Potential well**

Example 16. An electron is bound in a one-dimensional potential well of the width $1 \text{ \AA}$, but of infinite wall height. Find its energy values in the ground state, and also in the first two excited states.

Data: width of potential well, $a = 1 \text{ \AA} = 10^{-10} \text{ m}$

To find: Values of energy of the electron in the ground and the first two excited states

i.e., $E_1 = ?$, $E_2 = ?$, $E_3 = ?$,

Solution: The energy of the electron in 1D infinite potential well is,

$$E_n = \frac{n^2 h^2}{8ma^2}$$

where the values of E that we get for $n = 1, 2$ and 3 correspond to E_1 , E_2 and E_3

$$E_1 = \frac{n^2 h^2}{8ma^2} = \frac{(6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(10^{-10})^2}$$

$$E_1 = 6.0314 \times 10^{-18} \text{ J} = 37.64 \text{ eV}$$

$$E_2 = \frac{2^2 h^2}{8ma^2} = 2^2 \left[\frac{h^2}{8ma^2} \right] = 4 E_1$$

$$E_2 = 4 \times 6.0314 \times 10^{-18} \text{ J} = 150.54 \text{ eV}$$

$$E_3 = \frac{3^2 h^2}{8ma^2} = 3^2 \left[\frac{h^2}{8ma^2} \right] = 9 E_1$$

$$E_3 = 9 \times 6.0314 \times 10^{-18} \text{ J} = 338.7 \text{ eV}$$

Example 17. An electron is bound in a one-dimensional potential well of infinite height of width 0.12 nm . Find its energy values in the ground state, and also in the first two excited states.

Data: width, $a = 0.12 \text{ nm} = 0.12 \times 10^{-10} \text{ m}$

To find: Values of energy of the electron in the ground and the first two excited states

i.e., $E_1 = ?$, $E_2 = ?$, $E_3 = ?$,

Solution: The energy of the electron in 1D infinite potential well is,

$$E_n = \frac{n^2 h^2}{8ma^2}$$

for $n = 1, 2$ and 3 correspond to E_1 , E_2 and E_3

$$E_1 = \frac{n^2 h^2}{8ma^2} = \frac{(6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(0.12 \times 10^{-9})^2}$$

$$E_1 = 26.16 \text{ eV}$$

$$E_2 = 4 E_1 = 4 \times 26.16 = 104.61 \text{ eV}$$

$$E_3 = 9 E_1 = 9 \times 26.16 = 235.44 \text{ eV}$$

Example 18. An electron is trapped in a 1-dimensional potential well of infinite height of width 0.5 nm . If transition takes place from the first excited state to the ground state, then find the wavelength of photon emitted.

Data: width, $a = 0.5 \text{ nm} = 0.5 \times 10^{-10} \text{ m}$

To find: Wavelength of photon emitted, $\lambda = ?$

Solution: The energy of the electron in 1D infinite potential well is,

$$E_n = \frac{n^2 h^2}{8ma^2}$$

for $n = 1$, the ground state energy

$$E_1 = \frac{n^2 h^2}{8ma^2} = \frac{(6.625 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(0.5 \times 10^{-9})^2}$$

$$E_1 = 2.41 \times 10^{-19} = 1.507 \text{ eV}$$

for $n = 2$, the first excited state energy

$$E_2 = 4 E_1 = 4 \times 1.507 = 6.025 \text{ eV}$$

Energy difference,

$$\Delta E = E_2 - E_1 = 6.025 - 1.507 = 4.52 \text{ eV}$$

$$\Delta E = 7.23 \times 10^{-19} \text{ J}$$

Wavelength of photon emitted,

$$\lambda = \frac{hc}{\Delta E} = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{7.23 \times 10^{-19}}$$

$$\lambda = 274.8 \text{ nm}$$